Flow-induced vibrations in FOWTs – Floating Offshore Wind Turbines

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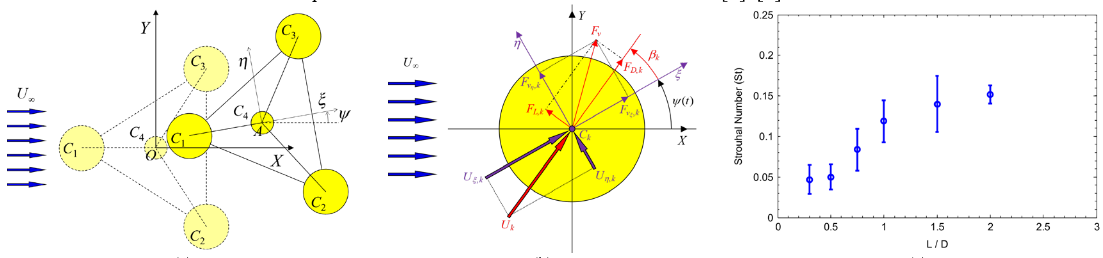
Extended Abstract

With its vast spatial availability and high energy potential, offshore wind represents a key opportunity in the renewable energy sector. Nevertheless, Floating Offshore Wind Turbines (FOWTs) face significant operational challenges, one of which is Vortex-Induced Motion (VIM). This phenomenon, a specific case of Vortex-Induced Vibration (VIV), arises when the vortices shedding from the columns synchronize with the moored platform natural frequencies, potentially compromising mooring lines and the wind turbine operation. To simulate VIM efficiently, we proposed a Reduced Order Model (ROM) based on wake oscillator dynamics, which offer a lightweight computational alternative to more intensive Computational Fluid Dynamics (CFD) simulations. A series of studies [1]–[4] has demonstrated the ROM effectiveness in capturing the behavior of multi-column FOWTs, showing close alignment with experimental observations [1], [3] and outperforming 2D CFD models in predictive accuracy [2]. Detailed formulations of these models are available in [1] and [3], where the floating structure is idealized as a rigid body constrained to horizontal-plane motion. Using Lagrange's equation formalism, the ROM was derived having three primary generalized coordinates associated with the FOWT rigid-body degrees of freedom (DoFs) on the horizontal plane. Hydrodynamic forces due to vortex shedding are represented via a heuristic approach: two wake oscillators, governed by forced van der Pol equations, are assigned per column, resulting in an augmented model with $3 + 2N$ DoF, where N denotes the number of columns. In the case illustrated in Fig. 1 (a), N equals 4, reflecting the existence of a central column. A full list of model parameters and their derivation can be found in [1]–[4].



(a) (b) (c)

Figure 1 – ROM framework. (a): Coordinate systems and definitions, with the platform initially aligned at 0-degree current heading, adapted from [1]; (b): Force diagram of the k-th column, illustrated for motion toward the left and downward direction, based on [1]; (c): Mean Strouhal number values and corresponding standard deviations as a function of cylinder aspect ratio, within the Reynolds number range 10,000 < Re < 50,000, from [5].

The resulting $3 + 2N$ DoF ROM is expressed as follows:

$$\tilde{\mathbf{M}}\ddot{\tilde{\mathbf{q}}} = \tilde{\mathbf{Q}}_c + \tilde{\mathbf{Q}}_{nc};\quad \tilde{\mathbf{q}} = \begin{bmatrix}\mathbf{q}\\ \mathbf{w}\end{bmatrix};\quad \tilde{\mathbf{M}} = \begin{bmatrix}\mathbf{M} & \mathbf{0}\\ \mathbf{A}_w & \mathbf{1}\end{bmatrix};\quad \tilde{\mathbf{Q}}_c = \begin{bmatrix}\mathbf{Q}^m - \mathbf{Q}^I\\ \mathbf{Q}^r_w\end{bmatrix};\quad \tilde{\mathbf{Q}}_{nc} = \begin{bmatrix}\mathbf{Q}^v + \mathbf{Q}^v_p\\ \mathbf{Q}^v_w\end{bmatrix}. \tag{1}$$

In this formulation, $\tilde{\mathbf{M}}$ and $\tilde{\mathbf{q}}$ denote the augmented inertia matrix and the generalized coordinate vector, respectively. On the right-hand side of the Eq. (1), $\tilde{\mathbf{Q}}_c$ and $\tilde{\mathbf{Q}}_{nc}$ represent the extended generalized force vectors. The first of these accounts for forces dependent on the system's configuration, while the second captures non-conservative effects. The vector $\mathbf{q}$ is a 3x1 array containing the generalized coordinates of the platform, i.e., X, Y, and ψ, describing its horizontal displacements and yaw angle. The wake dynamics are represented by $\mathbf{w}$, a $2N$x1 vector collecting the wake oscillator variables for each of the N columns. The platform's inertia is represented by matrix $\mathbf{M}$, of size 3x3, which includes both rigid body and added mass effects, evaluated as the free surface potential flow quantities at the asymptotic limit of zero-frequency. Coupling between columns' motion and the wake oscillators is captured by the matrix $\mathbf{A}_w$, which has dimensions $2N$x$2N$. Several generalized force vectors are defined: $\mathbf{Q}^m$, $\mathbf{Q}^I$, $\mathbf{Q}^v$, and $\mathbf{Q}^v_p$, all of size 3x1, representing mooring forces [6], inertial forces due to Coriolis and centrifugal effects, vortex-induced viscous forces (see Fig. 1 (b)), and pontoons' viscous drag effects, respectively. Additionally, $\mathbf{Q}^r_w$ and $\mathbf{Q}^v_w$, both $2N$x1 vectors, account for the restoring and damping forces acting on the wake oscillators. Each column k ($k = 1, \ldots, N$) is associated with a van der Pol based wake oscillator model defined in local coordinates aligned with the in-line (ξ) and transverse (η) directions (see Fig. 1 (b)), as described in Eq. (2):

$$\ddot{w}_{\xi,k} + \varepsilon_\xi\omega_{s,k}\left(w_{\xi,k}^2 - 1\right)\dot{w}_{\xi,k} + 4\omega_{s,k}^2 w_{\xi,k} = \frac{A_\xi}{D_k}a_{\xi,k};\quad \ddot{w}_{\eta,k} + \varepsilon_\eta\omega_{s,k}\left(w_{\eta,k}^2 - 1\right)\dot{w}_{\eta,k} + \omega_{s,k}^2 w_{\eta,k} = \frac{A_\eta}{D_k}a_{\eta,k}, \tag{2}$$

The variables $w_{\xi,k}$ and $w_{\eta,k}$ serve as internal generalized coordinates that phenomenologically capture the wake dynamics and its interaction with the platform. The parameters ε_ξ and ε_η denote damping coefficients, while A_ξ and A_η represent inertial coupling terms. Each column is characterized by a diameter D_k, and its accelerations along body-fixed directions are represented by $a_{\xi,k}$ and $a_{\eta,k}$. These acceleration components are used to produce forcing terms on the wake oscillators through the inertial coupling approach. The vortex shedding frequency, defined as $\omega_{s,k} = 2\pi S_{t,k}(U_k/D_k)$, depends on the Strouhal number $S_{t,k}$ for columns of low aspect ratio and the relative velocity U_k between the k-th column and the surrounding flow (see Fig. 1 (c)). It is important to note that the wake oscillator in the in-line direction oscillates at twice the frequency of its transverse counterpart. This reflects a common assumption based on VIV experimental findings, where hydrodynamic drag forces exhibit a dominant frequency that is twice that of lift-induced forces. For simulation purpose, the coefficients ε_ξ, ε_η, A_ξ and A_η have been tuned using experimental data and CFD results. Case studies are summarized to demonstrate the applicability of the ROM. The methodology is applied in [1] and [3] to reproduce towing tank experiments conducted with 1:72.72 and 1:100 small scale models of conceptual FOWTs. In [1], a triangular FOWT features a central column surrounded by three peripheral columns, interconnected by braces with circular cross-sections. In [3], a rectangular FOWT consists of four peripheral columns connected by pontoons with rectangular cross-sections. For both case, simulation results span over more than 30 reduced velocity values, defined by $V_R = U_\infty/(Df_{n,Y})$, where U_∞ is the freestream current speed, D is the diameter of the peripheral columns, and $f_{n,Y}$ is the transverse natural frequency. In [1], the range explored covers $3 < V_R < 24$. In [3], the range covers $3 < V_R < 30$. Both ranges correspond to Reynolds numbers between 8,000 and 70,000. Results from [1] and [3] are presented by Figures 2 (a–c) and (d–f) for non-dimensional amplitudes and horizontal-plane trajectory of the FOWT's center.

Results are rewarding, showing the ROM's capacity to capture the VIV nonlinear resonance in the transversal direction, at reduced velocities below 15 and 18- for the triangular and rectangular shaped FOWTs, respectively, whereas revealing a second resonance regime dominated by yawing at higher reduced velocities. Future work shall investigate extending the ROM to support towing simulations during the offshore installation phase of floating wind systems. Bifurcation maps as well as post-critical behaviors shall be studied, assessing their effects on VIM.

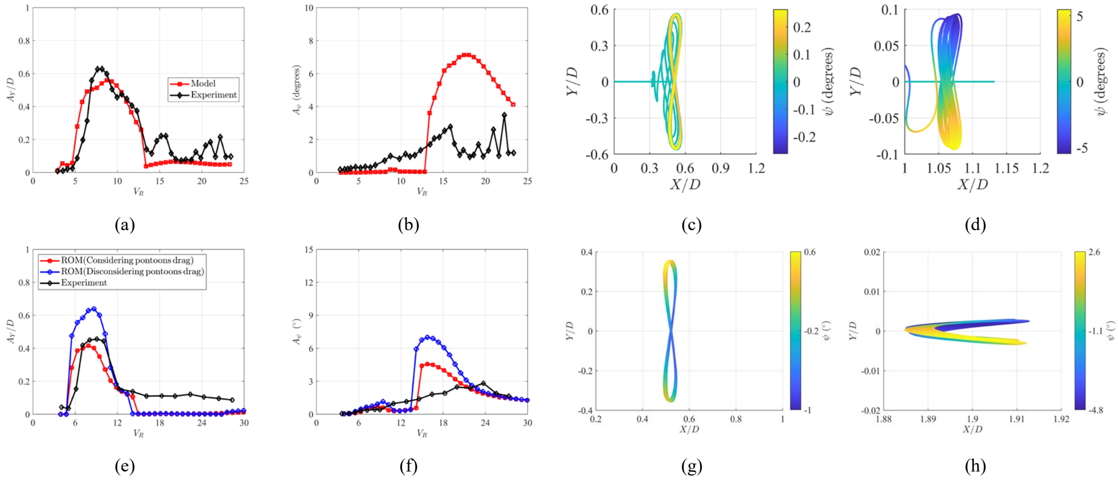


Figure 2 – FOWT dynamics on the horizontal plane for a current incidence of 0 degrees. Cross flow and yaw amplitudes for [1] in (a) & (b) and for [3] in (e) & (f). FOWTs' center trajectory, with yaw in color scale for [1] in (c): $V_R = 8.73$ & (d): $V_R = 16.87$ and for [3] with pontoons drag in (g): $V_R = 9.45$ & (h): $V_R = 18.89$.

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